AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 3 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 3 of 8

20 marks · about 25 minutes

Key information

- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$
- Pure water is neutral at every temperature because $[H^+] = [OH^-]$; its pH is only 7 at 25 °C
- Weak acid on its own: $[H^+] = \sqrt{(K_a \times [HA])}$; $K_a = 10^{-pKa}$
- Indicator: its colour-change range must lie within the vertical section of the pH curve

Section A: Short-answer questions

01 2,4,6-Trichlorophenol is a weak monoprotic acid, with $K_a = 2.51 \times 10^{-8}$ mol dm^{-3} at 298 K.

What is the concentration, in mol dm^{-3}, of hydrogen ions in a 2.00×10^{-3} mol dm^{-3} solution of 2,4,6-trichlorophenol at 298 K?

[1 mark]

A 5.02×10^{-11}

B 7.09×10^{-6}

C 1.26×10^{-5}

D 3.54×10^{-3}

02 State the meaning of the term *Brønsted–Lowry acid*.

[1 mark]

03 Which indicator should be used in a titration to find the concentration of a solution of methylamine using 0.010 mol dm^{-3} hydrochloric acid?

[1 mark]

A Thymol blue (pH range 1.2–2.8)

B Bromophenol blue (pH range 3.0–4.6)

C Phenol red (pH range 6.8–8.4)

D Phenolphthalein (pH range 8.3–10.0)

04 At 50 °C, $K_w = 5.48 \times 10^{-14}$ mol^2 dm^{-6}. Calculate the pH of a 0.0100 mol dm^{-3} solution of sodium hydroxide at 50 °C. Give your answer to 2 decimal places.

[2 marks]

05 Equal volumes of pairs of solutions are mixed. Which pair forms a buffer solution?

[1 mark]

A ammonia and ammonium chloride

B ammonia and methylamine

C ethanoic acid and methanoic acid

D hydrochloric acid and sodium hydroxide

06 A weak acid with $K_a = 1.35 \times 10^{-5}$ mol dm^{-3} is titrated with sodium hydroxide. Calculate the pH when exactly half of the acid has been neutralised.

[1 mark]

Section B: Longer questions

07 For butanoic acid, $K_a = 1.51 \times 10^{-5}$ mol dm^{-3}

07.1 20.0 cm^3 of 0.100 mol dm^{-3} sodium hydroxide solution are added to 25.0 cm^3 of 0.100 mol dm^{-3} butanoic acid solution.

Calculate the pH of the solution formed.

[5 marks]

07.2 A student plans to titrate butanoic acid solution with a solution of ethylamine. Explain why this titration could not be done using an indicator.

[2 marks]

08 Pure water dissociates slightly: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$ $\Delta H = +57$ kJ mol^{-1}

Table 1 shows how K_w varies with temperature.

Temperature / °C	K_w / mol^2 dm^{-6}
10	2.93×10^{-15}
20	6.81×10^{-15}
25	1.00×10^{-14}
30	1.47×10^{-14}
50	5.48×10^{-14}

08.1 Explain why the value of K_w increases as the temperature increases.

[2 marks]

08.2 Give the expression for pH. Calculate the pH of pure water at 50 °C. Give your answer to 2 decimal places. Explain why water is neutral at 50 °C.

[4 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Sheet 3 of 8

Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7405/3 Jun 2017 Q09*	1
02	Proton donor / H^+ donor	*Source: Practice question (written for this set)*	1
03	**B**	*Source: 7405/3 Jun 2018 Q32*	1
04	M1 $[H^+] = 5.48 \times 10^{-14} \div 0.0100 = 5.48 \times 10^{-12}$ M2 pH = 11.26	*Source: Practice question (written for this set)* Using $K_w = 1.00 \times 10^{-14}$ (pH 12.00) scores M1 only if method shown	2
05	**A**	*Source: 7405/3 Jun 2023 Q20*	1
06	4.87 (pH = pK_a at half-neutralisation)	*Source: Practice question (written for this set)*	1
07.1	M1 initial n(butanoic acid) = 2.5×10^{-3} mol M2 n(NaOH) = 2.0×10^{-3} mol M3 final n(acid) = $2.5 \times 10^{-3} - 2.0 \times 10^{-3} = 5.0 \times 10^{-4}$ mol M4 $[H^+] = K_a \times [HX] \div [X^-] = 1.51 \times 10^{-5} \times 5.0 \times 10^{-4} \div 2.0 \times 10^{-3}$ ($= 3.775 \times 10^{-6}$) M5 pH = 5.42	*Source: 7405/1 Jun 2023 Q10.4–10.5* M4 allow volumes cancelled out M5 dependent on a correct expression for $[H^+]$ in M4	5
07.2	M1 weak acid and weak base / alkali titration M2 pH change too gradual / not sharp at the equivalence point	M2 allow no vertical / steep section on pH curve	2
08.1	M1 equilibrium is endothermic (in forward direction) M2 equilibrium shifts to the right to oppose / minimise the temperature increase	*Source: 7405/1 Nov 2021 Q06.2–06.3* Ignore more H^+ and OH^- formed	2
08.2	M1 pH = $-\log_{10}[H^+]$ M2 $[H^+] = \sqrt{(5.48 \times 10^{-14})}$ ($= 2.34 \times 10^{-7}$) M3 pH = 6.63 M4 $[H^+] = [OH^-]$	M4 allow equal amounts of H^+ and OH^-	4